Small Area Estimation with the Fay-Herriot Model

Fall 2026
13.11.2026

University of Colorado Denver

Brady Lamson
brady.lamson@ucdenver.edu

Contents

1. Introduction 1
1.1. Background 1
2. The Data 1
2.1. Data Sources 1
2.1.1. 2020 Decennial Census of Population and Housing 1
2.1.2. American Community Survey 1
2.1.3. NASS Crop Data Layer 2
2.2. Data Processing 2
2.2.1. American Community Survey 2
2.2.1.1. Logit Transformation of the Response 3
2.2.1.2. Logit Transformation of the Standard Error 3
2.2.2. 2020 Decennial Census of Population and Housing 4
2.2.3. NASS Crop Data Layer 5
2.3. Data Analysis 5
2.3.1. Iowa Spatial Structure 5
2.3.2. 2020 Decennial Census of Population and Housing 6
2.3.3. American Community Survey 7
2.3.4. Tests of Spatial Autocorrelation for Housing Vacancy Rates 10
2.3.5. NASS Crop Data Layer 10
3. Methods 13
3.1. Fay-Herriot Model 13
3.2. Bayesian Fay-Herriot Model 14
3.2.1. Prior Distributions 14
3.2.2. Full Posterior Distribution 15
3.3. Spatial Random Effects 15
3.4. Auxillary covariates 16
4. Results 16
4.1. Summary Results 16
4.1.1. Mean Squared Error and Mean Absolute Error 17
4.1.2. Mean Standard Deviation and Mean Interval Width 17
4.1.3. Interval Coverage 18
4.1.4. Continuous Ranked Probability Score (CRPS) 18
4.2. Model Diagnostics 18
4.2.1. Summary Diagnostics 18
4.2.2. Trace Plots and Paired Scatterplots 19
4.3. Model Coefficients 21
4.4. Estimation Calculation Examples 22
4.4.1. Intercept Only Model Frequentist 22
4.4.2. Intercept Only Model Bayesian 23
4.4.3. ICAR Model 23
4.5. Model Estimate comparisons 24
4.5.1. Residuals and Standard Deviations 24
4.5.2. Interval Coverage Outcomes 26
5. Conclusion 27
5.1. Limitations 27
5.1.1. Spatial Covariate 27
5.1.2. Decennial Census as source of truth 28

5.1.3. Neighbor matrix encoding 28
5.1.4. Use of abstracted modeling packages 29
5.2. Discussion 29
Bibliography 29
6. Appendix 31
6.1. EMDI and BRMS Code 31
6.1.1. Emdi Intercept Only 31
6.1.2. BRMS Intercept Only 31
6.1.3. BRMS ICAR 31
6.2. STAN Code 31
6.2.1. BRMS Intercept Only STAN Code 31
6.2.2. BRMS ICAR STAN Code 33

1. Introduction

1.1. Background

Much of the data that statisticians rely on for their work is taken for granted.

2. The Data

2.1. Data Sources

2.1.1. 2020 Decennial Census of Population and Housing

The first source of data for this project comes from the 2020 Decennial Census. Mandated by the Constitution of the United States, the census is carried out every 10 years by the U.S. Census Bureau and "determines the number of people living in the United States" [1]. Though the primary purpose of the census is to distribute seats in the U.S. House of Representatives [1], it is also the official source of high level demographic and housing data for the entire country. As such, it is used as the source of truth in this project. Every single model will be compared to the values provided by the 2020 Decennial Census to evaluate the quality of predictions. It is important to note that, though treated as the source of truth, that the values provided by the U.S. census are not perfect. An analysis performed by the U.S. Government Accountability Office in 2024 found 14 states exhibiting statistically significant net coverage errors in the 2020 census, though Iowa is absent from this list [2]. Accounting for this uncertainty is outside the scope of this project, so all values provided by the census are treated as the true values.

2.1.2. American Community Survey

The second, and primary source of data for this project comes from the American Community Survey (ACS). The ACS is an ongoing survey performed by the U.S. Census Bureau about the population of the United States of America [3]. The Census Bureau is primarily known for the U.S. Decennial Census, but the lesser known ACS serves as a crucial companion piece to it. Instead of every 10 years, the ACS is sent out to a sample of addresses every single month of every single year [4]. The ACS in this aspect fills a very important role, as it is able to provide current information between decennial census collections, and is also able to cover a more robust set of topics than the census. The tradeoff is that the ACS comes from a sample of households, so every result is met with a certain degree of uncertainty. However, unlike traditional surveys, individuals selected for the ACS are legally obligated to respond to it [5]. The consequence of this is that the uncertainty in the ACS results can be treated as a known quantity, which allows for mathematical techniques that are not normally possible with traditional surveys that are optional to respond to.

The data from the ACS comes in two primary forms, 1-year and 5-year estimates. The 1-year estimates are less precise but more current, encompassing 12 months of collected data but only providing information on areas with a population size of at least $65,000$ [6]. The 5-year estimates are based on 60 years of collected data, and so are more precise but less

current. The 5-year estimates also provide information on all areas surveyed, so this even includes extremely sparsely populated county or census tract level data [6].

This project will be using the 2020 5-year estimates, encompassing survey data from 2016-2020 for each county in Iowa. The variables used are `B25002_001` and `B25002_003` which correspond to the number of total housing units and vacant housing units respectively. The ratio of the two is used to calculate a vacancy rate for each county in Iowa which functions as the response for this project.

2.1.3. NASS Crop Data Layer

The third source of data for this project is known as the Crop Data Layer, or CDL. According to the USGS, the CDL, created by the USDA National Agricultural Statistics Service (NASS), is "an annual raster, geo-referenced, crop-specific land cover data layer produced using satellite imagery and extensive agricultural ground reference data" [7]. The USDA website carries direct downloads to every annual data layer going back to 2008. The year used for this analysis is the same as the decennial census, 2020. This dataset contains per-pixel cropland classifications at a 30-meter resolution for the entire country. The classifications were done using a machine learning model trained primarily on the Farm Service Agency (FSA) Common Land Unit (CLU) data, though some states incorporate additional auxiliary data from other agencies and institutions [7]. This data within the region of Iowa is used to create the spatial covariate for the Fay-Herriot model. Incorporation of pixel classification uncertainty in the CDL is outside the scope of this project, so all classifications are treated as the true classifications.

2.2. Data Processing

2.2.1. American Community Survey

ACS data was acquired for this project through the Census Bureau API, indirectly interfaced with via the `tidycensus` R package on CRAN. Data was pulled from the 2020 5-year ACS estimates for the `B25002_001` and `B25002_003` variables, which as previously mentioned correspond to the total housing units and vacant housing units per county. The resulting dataframe has a grain of one row per variable per county, so 2 rows per county. Each row contains the estimate for that variable, its corresponding Margin of Error (MoE) and the spatial geometry of that county. This dataframe was pivoted to a grain of one row per county to facilitate creation of the response variable, vacancy rates. For each county, the vacant unit count was divided by the total housing unit count to get the vacancy rate of that county.

$$\text{Vacancy Rate} = \left(\frac{\text{Vacant Housing Units}}{\text{Total Housing Units}}\right) \tag{1}$$

To get the standard deviation of this new estimate, two steps are necessary. First, a conversion from MoE to standard error is required for each estimate. Then, the standard errors of both estimates must be combined to form the standard error of the proportion. The Census Bureau

provides guidelines on both steps, and the same document will be referred to for both [8]. Conversion from MoE to SE is done by simplying dividing the MoE by 1.645.

$$\text{Standard Error} = \frac{\text{Margin of Error}}{1.645} \tag{2}$$

As for the standard error of the proportion response, the following setup is used. First, let $P = \frac{A}{B}$, where A is a subset of B. A will represent the vacant housing units, B will be represent all housing units. Then, the standard error of a counties housing vacancy rate can be represented as

$$\text{SE}(P) = \frac{1}{B}\sqrt{\text{SE}(A)^2 - P^2 \times \text{SE}(B)^2} \tag{3}$$

2.2.1.1. Logit Transformation of the Response

As the Fay-Herriot model relies on linear regression as a key component, a proportion response can behave poorly. The domain of the output of a linear reression is the entire real line, whereas a proportion is bounded between 0 and 1. In a more traditional scenario this would lead to a logistic regression model, though usage of the Fay-Herriot model requires alternative methods to overcome this limitation. Logit transformations have been used in this scenario before, such as in research performed by Cucu Sumarni et al. that found a "logit transformation within the SAE framework improved the precision of proportion estimates" [9]. This takes a response bounded between 0 and 1 and maps it to the entire real line. Thus, our transformed response can be represented as

$$Q = \ln\left(\frac{P}{1-P}\right) \tag{4}$$

2.2.1.2. Logit Transformation of the Standard Error

Yet again, additional work is required for the conversion of the standard error. For this, the delta method was utilized. For reference,

Definition 2.2.1.2.1 (Delta Method)**.** Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \to N(0, \sigma^2)$ in distribution. For a given function g and a specific value of θ, suppose that $g'(\theta)$ exists and is nonzero. Then

$$\sqrt{n}[g(Y_n) - g(\theta)] \to N\left(0, \sigma^2[g'(\theta)]^2\right) \tag{5}$$

in distribution.

To apply this, let

$$g(P) = \ln\left(\frac{P}{1-P}\right) \tag{6}$$

Then,

$$g'(P) = \frac{1}{P - P^2} \tag{7}$$

We can then substitute this into the definition to get the updated variance, σ_Q^2.

$$\begin{aligned}\sigma_Q^2 &= \sigma^2[g'(P)]^2) \\ &= \sigma^2\left(\frac{1}{P-P^2}\right)^2 \\ &= \frac{\sigma^2}{(P-P^2)^2} \\ &= \frac{\sigma^2}{P^2(1-P)^2}\end{aligned} \tag{8}$$

From this, our transformed standard error is

$$\sigma_Q = \frac{\sigma}{P(1-P)} \tag{9}$$

The final dataframe contains a variety of other columns for diagnostic and exploratory purposes, but the dataframe of modeling components looks like so:

County	Vacancy Rate	SE	Vacancy Rate (Logit)	SE (Logit)	Geometry
Polk	0.064	0.003	−2.69	0.05	-
Grundy	0.076	0.011	−2.50	0.154	-
-	-	-	-	-	-

Table 1: Snippet of ACS Dataframe used for modeling. Untransformed columns and geometry included for the sake of reference and demonstration.

2.2.2. 2020 Decennial Census of Population and Housing

Ingestion and processing of the 2020 decennial census data was handled in a similar manner to the ACS data. Data was acquired through the `tidycensus` package, used to access the 2020 Census Demographic and Housing Characteristics File (DHC). The variables pulled were `H3_001N` and `H3_003N`, which correspond to total housing units and total vacant units repsectively. The grain of the data is the same as the original ACS data, one row per county per variable. The final table had one row per county, and a new column representing the housing vacancy rate was created in percent units. These values were not passed through a LOGIT transformation as they are intended for comparison against the predictions after they have been transformed back into the original proportion scale.

To ensure the accuracy of census housing vacancy rates, output was compared to the official housing vacancy numbers provided by the Census Bureau in their 2020 Iowa breakdown [10]. All of the values matched their values presented on that sites dashboard. Below is a snippet of the data.

County	Total Housing Units	Vacant Housing Units	Vacancy Rate
Polk	210184	13293	0.06
Grundy	5465	396	0.07
-	-	-	-

Table 2: Snippet of the Census DataFrame containing the true values for model evaluation. Non-proportion columns retained for the sake of reference and demonstration.

2.2.3. NASS Crop Data Layer

The crop data layer requires careful processing to be usable in this project. At a 30×30 meter resolution, the 2020 crop data layer raster image has a rectangular dimension of 96523×153811 pixels, or 14.8 billion pixels total. About 6.2 billion (42%) pixels are empty background pixels and 8.7 billion encompassing actual categorizations. Naive calculations performed on this object are prone to overloading the memory of the machine running them. The `terra` package was used for loading and processing this file. For filtering down to only Iowa's portion of the CDL, the `tigris` package was utilized to download Iowa's TIGER shapefiles from the U.S. Census. The coordinate reference systems (CRS) of the two objects were aligned to NAD83 Conus Albers (ESPG:5070). Cropping and masking is then performed on the CDL using the Iowa shapefile object. Cropping filters the CDL down to the rectangular region containing Iowa, and masking changes all pixels outside of the state borders to NA.

This new object, the Iowa CDL, contains a rectangular dimension of 11671×17794, or approximately 207 million pixels total. That is 1.3% the size of the original CDL. Of those, about 46 million (22%) are empty background, and appoximately 162 million (78%) pixels are categorized. From here, the goal is to get a table containing the frequency and proportion breakdown of every county in the state. To accomplish this goal, a similar process was used as before. County shapefiles were acquired using `tigris` and, using an iterative approach over the counties that leverages cropping and masking, the frequencies of every CDL classification were captured alongside the total number of pixels within the county's borders. A dataframe was created from this that contains one row per county, with the percent of the pixels in the county that were classified as either corn or soybeans. The proportion from this functions as the spatial covariate for the model and can be represented as:

$$\text{Proportion Corn or Soy}_i = \frac{\text{\# of Corn Pixels}_i + \text{\# of Soybean Pixels}_i}{\text{\# of Pixels Total}_i} \tag{10}$$

With i representing the index of a given county.

Below is a snippet of the resulting dataframe. Values are saved as percentages and converted to proportions at the modeling stage.

County	Soybeans (%)	Corn (%)	Corn or Soy (%)
Polk	17.8	22.4	40.2
Grundy	36.0	52.5	88.5
-	-	-	-

Table 3: Snippet of the CDL DataFrame containing the spatial covariate of the model. The individual soy and corn columns are kept for reference.

2.3. Data Analysis

2.3.1. Iowa Spatial Structure

Iowa was chosen for this project for a few reasons. The first is that it has some very populated counties and very sparsely populated counties. This allows for a good demonstration of the

Fay-Herriot model as it can effectively pool from informative counties. This will be discussed in more detail in later sections.

Second is the layout of its counties. Iowa has 99 counties total and they are arranged in a very neat grid as shown in the figure below. On average, each county has 5.9 neighbors with a minimum of 3 and a maximum of 8. There are no disjointed islands or bizarre geometries that would complicate spatial calculations as well. As such, this state shows useful properties for both Fay-Herriot modeling and spatial methods.

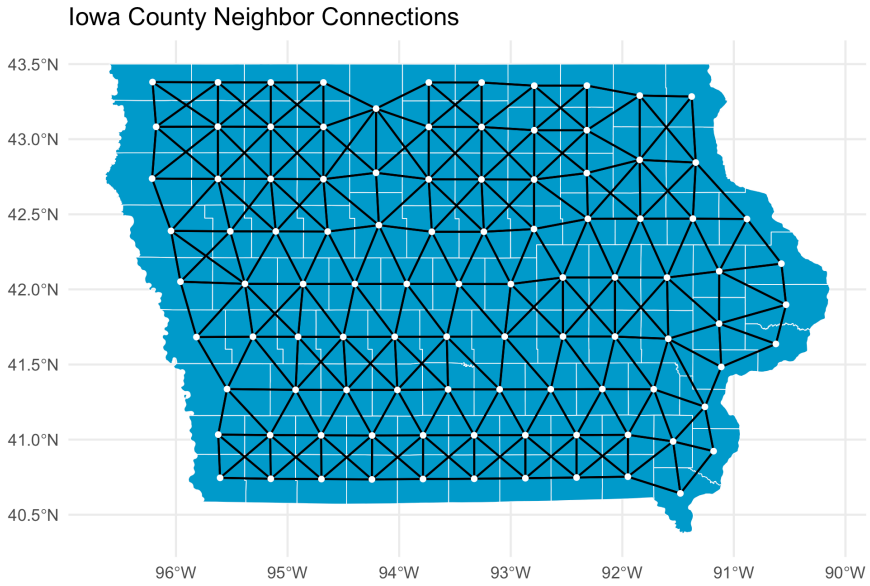


Figure 1: Plot showing adjacent county links across Iowa.

2.3.2. 2020 Decennial Census of Population and Housing

The counties in Iowa vary dramatically in terms of their population and housing. The number of total housing units can range from as little as 1,888 (Adams County) all the way up to 210,184 (Polk County). 29 of the 99 total counties have less than 5,000 total housing units.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	1888	274	0.05950
1st Quartile	4684	518	0.07892
Median	7037	701	0.10144
Mean	14271	1255	0.10842
3rd Quartile	10947	1131	0.11803
Maximum	210184	13293	0.41743

Table 4: Summary of the housing data from the 2020 decennial census. (N=99)

The same trend is seen with vacant units, as one would expect places with more houses also have more vacant houses. These variables show an almost perfect correlation ($r = 0.947$). This relationship does not hold however for vacancy rates which show a mild negative correlation with total housing units ($r = -0.237$). The highest vacancy rates appear on counties

with low total housing units. This is unsurprising as each vacant house contributes more to the vacancy rate than one would in larger counties.

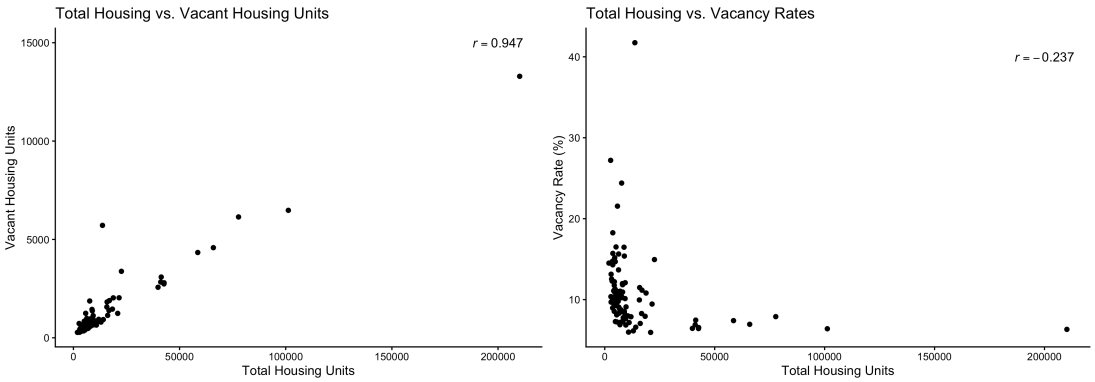


Figure 2: Plots showing the change of relationship when moving from pure vacant housing units to a vacancy rate.

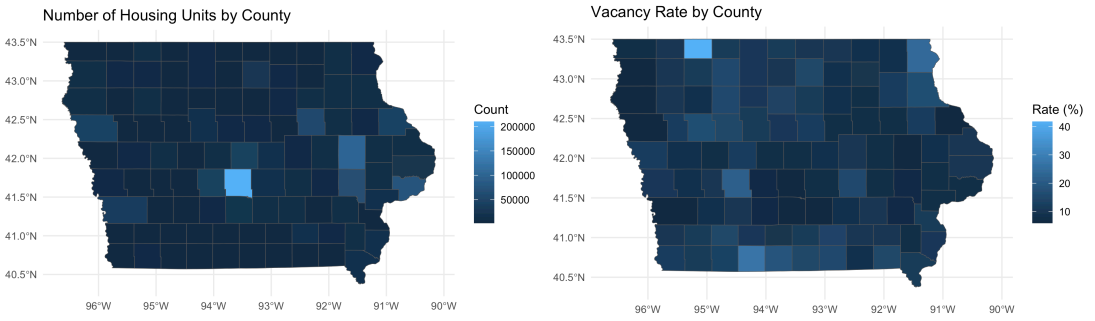


Figure 3: Spatial plots showing the distribution of housing units and vacancy rates by county across the state.

Examining these values at the spatial level, the overall population of housing units appears heavily concentrated in a small number of counties scattered throughout the state. In contract, vacancy rates show a far more mixed distribution of values. Visually it is unclear if there is evidence of spatial autocorrelation for the vacancy rates, but there does appear to be some grouping.

2.3.3. American Community Survey

Analysis begins with the ACS equivalent of Table 4, shown in Table 5. What is shown are near identical values, though there is a slight difference. The minimums never reach the same lows, and the maximums never reach the same highs. However, the values throughout the 1st and 3rd quartiles are consistently above those from the census.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	2011	296	0.0500
1st Quartile	4934	610.5	0.0900
Median	7199	776	0.1069
Mean	14220	1352.3	0.1210
3rd Quartile	11048	1370	0.1422
Maximum	204110	12985	0.3964

Table 5: Summary of the housing data estimates from the ACS. (N=99)

Examining the summary of the errors in Table 6 shows some extremely interesting behavior. The standard errors of the vacancy estimates are far higher than the total unit estimates. On average, the ratio of the vacant standard error to the total unit standard error is around 2.8, and goes all the way up to 7.1. This is before even accounting for the fact that the vacant unit estimates are far smaller, being a subset of the total. What this means is the vacancy unit estimates are far less precise than the total estimates are, and this lack of precision carries to the proportion as well. The relative standard errors for the vacancy estimates appear nearly identical to those of the vacancy rates.

Metric	Total SE	Vacant SE	Rate SE	Relative Total SE	Relative Vacant SE	Relative Rate SE
Minimum	14.59	43.77	0.003	0.000959	0.02784	0.02782
Q1	27.36	79.64	0.011	0.003855	0.10164	0.10164
Median	42.55	103.34	0.011	0.005198	0.12230	0.12230
Mean	50.85	129.06	0.014	0.005782	0.12231	0.12231
Q3	60.49	142.55	0.017	0.006975	0.14533	0.14532
Maximum	265.65	618.24	0.027	0.024768	0.25127	0.25127

Table 6: Summary of the estimate standard errors and relative standard errors from the ACS. The relative columns are calculated using (SE / Estimate) for each given estimate. Some adjustments to metric and column names made to preserve space.

Examining the relationships between the estimates, their corresponding standard errors and relative standard errors in Figure 4 shows us a few things. Firstly, it shows that the magnitude of standard errors tend to rise alongside the magnitude of the estimates themselves, and that this is consistent across all 3 types of estimates. Second, it shows that the opposite relationship arises when the standard errors are instead looked at relative to the size of the estimates themselves. In other words, larger estimates tend to be more precise. The counties with less housing units have larger errors in proportion to their estimates. This intutively makes sense as the samples themselves are smaller, there is less data to go off of for the estimates. It is worth noting that this does not mean that the larger estimates are more accurate, as no comparison has been made yet to the true values from the census.

The takeaway from these findings is that the smaller counties are going to be worth looking at closely in the model comparisons. In theory, these are the counties with the most

room for growth as they can pull from the higher precision in other counties with more housing units.

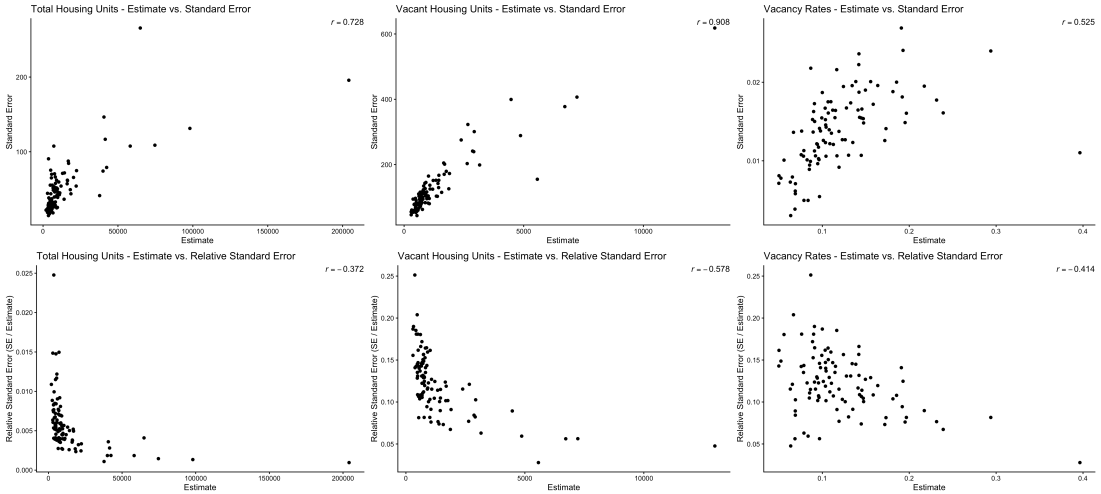


Figure 4: Plots showing the relationship estimates share with their standard errors and relative standard errors (SE / Estimate).

The distributions of the response and errors can be seen in Figure 5. The untransformed response is unimodal with a long right tail with a small bump in density around 0.4. The transformed response is very similar in distribution, but with a slightly less extreme right tail. Overall, it looks slightly more normally distributed than the untransformed response.

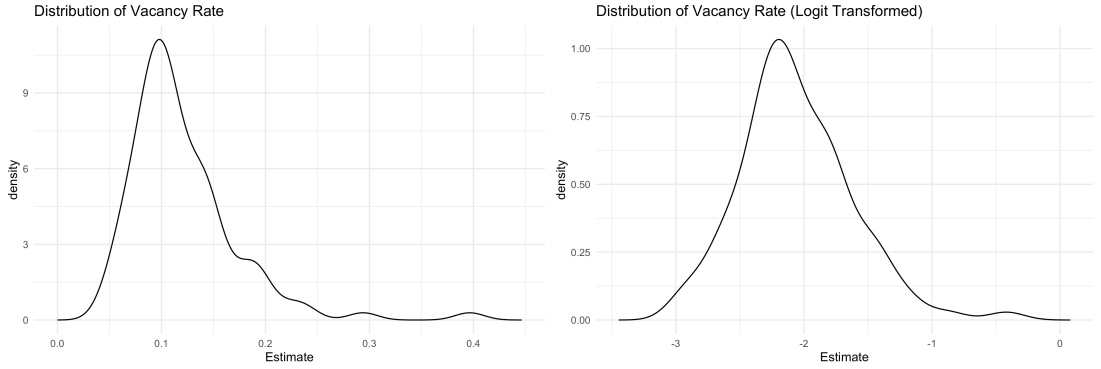


Figure 5: Distribution of the response before and after the logit transformation.

As for the errors shown in Figure 6, the shape stays mostly the same both before and after the response is transformed. The errors of the logit transformed response do show a couple additional bumps that aren't present in the original errors, though that may be more of an artifact of the plotting methodology than a meaningful change in distribution.

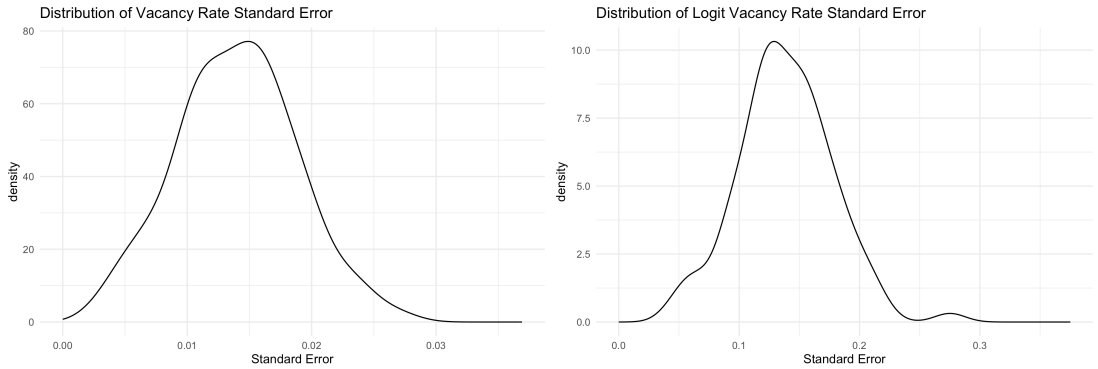


Figure 6: Distribution of the standard errors before and after the logit transformation of the response.

2.3.4. Tests of Spatial Autocorrelation for Housing Vacancy Rates

To assess the utility of an ICAR component in the Fay-Herriot model, Moran's I tests were performed on both the ACS response and the Census response separately to test for spatial autocorrelation. For this, the `moran.test` function was used from the `spdep` package on CRAN. For each dataset, two tests were performed under the constant risk hypothesis. One test uses binary neighbor encoding, another uses row standardization. The hypotheses for these tests are as follows:

$$\begin{aligned} H_0 &: \rho = 0 \\ H_a &: \rho > 0 \end{aligned} \tag{11}$$

From the results in the figure below, both forms of encoding from both datasets reach the same conclusions at the $\alpha = 0.05$ level. There does seem to be evidence of positive autocorrelation of housing vacancy rates under the constant risk hypothesis.

Data	Matrix Encoding	Moran's I	Expected I	Variance	Z-Score	p-value
ACS	Binary	0.147	−0.01	0.0034	2.79	0.0026
ACS	Row Standardized	0.160	−0.01	0.0034	2.94	0.0016
Census	Binary	0.130	−0.01	0.0034	2.48	0.0066
Census	Row Standardized	0.142	−0.01	0.0034	2.62	0.0043

Table 7: Moran's I results from both the ACS and Census datasets. These tests performed under the normality assumption and constant risk hypothesis.

It is worth noting that these tests were primarily used as a diagnostic and a quick check of method feasibility. The risk from incorrectly rejecting the null hypothesis is low, and so additional tests were not performed. What these results show is that an ICAR component in the Fay-Herriot model does have some grounding, it isn't being applied to a response with no evidence of spatial autocorrelation.

2.3.5. NASS Crop Data Layer

To help motivate many of the decisions made in Section 2.2.2, analysis will begin with the crop data layer itself. Examining Table 8 shows that the makeup of land in Iowa is very different from that of the USA as a whole. Corn and Soybeans make up 38% and 27% of

the states land coverage respectively. Adding those up, approximately 65% of Iowas land coverage is corn or soybeans. Neither of these even reach 5% of the land coverage in the USA overall. The plots shown in Figure 7 visualize the similar spatial distributions and land coverage of corn and soybeans across the state. Their similarities and high prevelance, alongside their importance to Iowas overall agriculture gave the motivation for using them as a proxy for agricultural coverage per county.

USA		Iowa	
Crop	**Percentage**	**Crop**	**Percentage**
Shrubland	23.88	Corn	38.29
Grassland/Pasture	16.24	Soybeans	26.52
Evergreen Forest	12.58	Grassland/Pasture	13.73
Deciduous Forest	10.93	Deciduous Forest	8.43
Corn	4.85	Developed/Open Space	3.52
Woody Wetlands	4.59	Developed/Low Intensity	1.79
Soybeans	4.39	Herbaceous Wetlands	1.34

Table 8: Top crops by percentage of land classification for both the full USA CDL and Iowa subset. Top 7 rows.

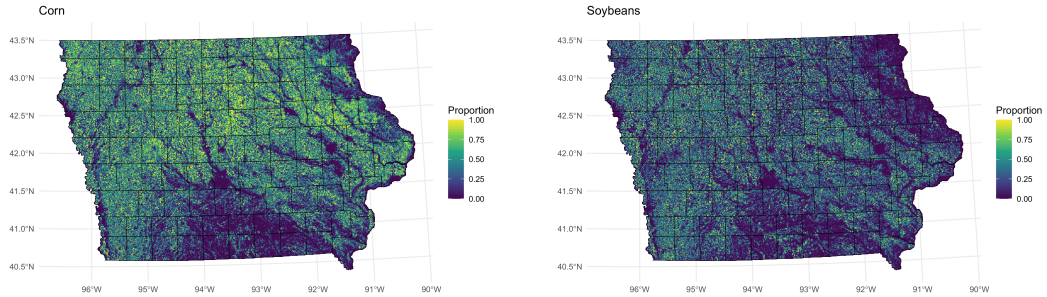


Figure 7: Plots showcasing the aggregated spatial distribution of corn and soybeans. Aggregation performed on each crop separately taking the mean of every 25x25 pixel area.

Amongst the various CDL classifications, there are ones that are referred to as "double crops". These are classifications with two different crops in them. There are 6 of these classifications total, examples including values like corn/soybeans and soybeans/oats. These were considered for inclusion in the covariate, though upon further inspection they were far too infrequent to justify. All 6 values total up to only around $4,000$ pixels total across all of Iowa. The max percentage for a double crop in a county was 0.042%. As such, these were ignored in all later work.

The top crops were examined for all counties to check for any other crops worth consideration. The top 3 crops for all 99 counties simplify down to only 6 cdl classifications. Those being corn, soybeans, developed/low intensity, developed/open space, grassland/pasture and deciduous forest. Nothing here appeared to be worth consideration for inclusion, so the decision was made to stick exclusively to corn and soybeans.

Narrowing down to the key crops at the county level, the overview in Table 9 shows that even the minimums show a large percentage of land coverage for the two key crops. At the most extreme almost 90% of land is classified as either corn or soybeans.

Statistic	Corn (%)	Soybeans (%)	Corn or Soybeans (%)
Minimum	8.58	6.53	19.95
1st Quartile	29.53	21.14	52.51
Median	41.32	27.47	69.36
Mean	37.73	26.42	64.15
3rd Quartile	47.73	32.22	77.84
Maximum	57.5	38.46	88.51

Table 9: Summary of the crop distribution across all Iowa counties.

Visualizing this information in Figure 8 shows some indication of spatial autocorrelation of land coverage. Many of the highest coverage counties are grouped together near the north west part of the state, and the lowest coverage counties are grouped in the south.

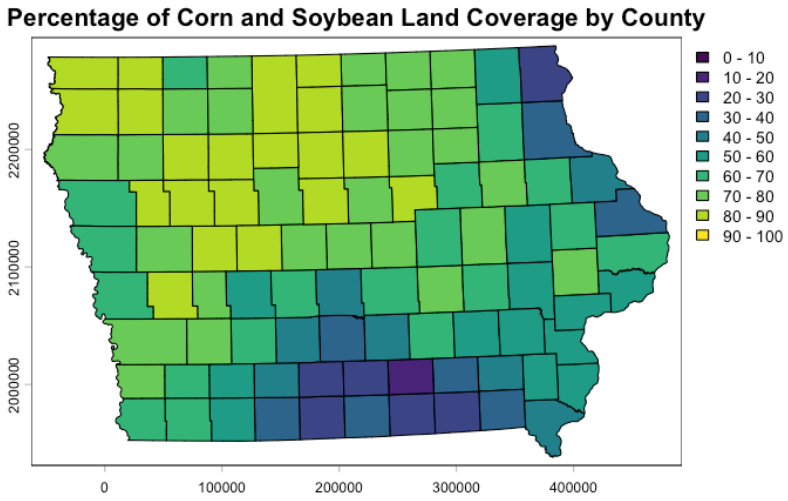


Figure 8: Plot of land coverage per county.

The plot in Figure 9 showcases a big problem with this choice of covariate. It appears to to show a minor negative relationship with vacancy rates, though it's not a convincing trend. The points on the plot appear very randomly scattered, with no obvious trend to point to. This plot begs the question of why this is even a covariate for the model. This is something that will be explained in more detail in the limitations section, however to keep it brief this is the result of a change in topic partway through the project. The previous response had a far stronger relationship with the crop data layer. Due to the extensive amount of work done on this dataset, it is kept in this report primarily as a showcase of methodology.

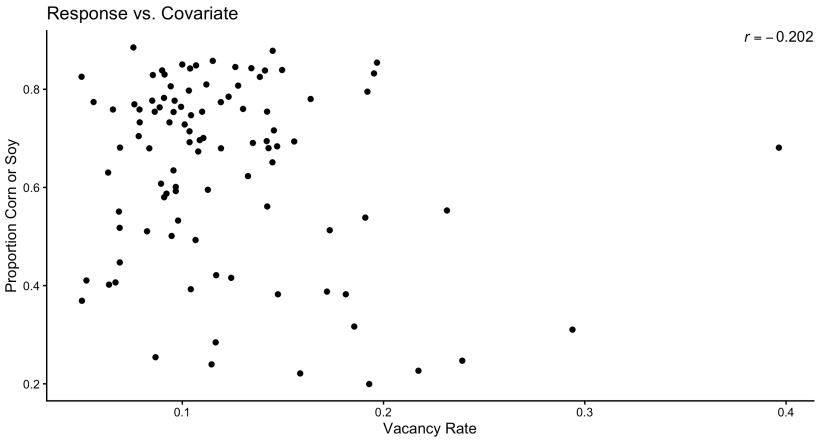


Figure 9: Scatterplot showing the a mild negative relationship between vacancy rates and corn/soy land coverage.

3. Methods

3.1. Fay-Herriot Model

The traditional Fay-Herriot model is, for $i = 1, \ldots, n$, defined as

$$Y_i = \theta_i + \varepsilon_i \tag{12}$$

$$\theta_i = \beta_0 + \boldsymbol{x_i^T \beta_x} + \mu_i \tag{13}$$

where $\varepsilon_i \sim N(0, \sigma_i^2)$ and $\mu_i \sim N(0, \sigma_\mu^2)$. Notation taken from previous work done by Porter et al. [11]. These error terms, ε_i and μ_i are mututally indepdendent. θ_i represents the population parameter of interest. In this project's case that is the true housing vacancy rate for county i. Y_i represents the direct estimate of θ_i, these are the estimated vacancy rates from the ACS. The variance of $\varepsilon_i, \sigma_i^2$, is assumed known. These are the variances calculated from each estimate's margin of error provided by the ACS. The other error term μ_i represents the area level random effects. The auxillary information is given by the q-dimensional vector $\boldsymbol{x_i}$ with corresponding q-dimensional regression parameters $\boldsymbol{\beta_x}$ and intercept β_0 [11].

This model combines the direct estimates and regression-based fitted values taken from the overall random effects and area level sampling variances. Each area's sampling variance σ_i^2 is used to calculate a corresponding shrinkage value γ_i which determines how much the Fay-Herriot estimate leans towards the original direct estimate or regression fitted value.

For the intercept-only model, the shrinkage value can be represented as

$$\hat{\theta}_i^{\text{FH}} = \hat{\gamma}_i \hat{Y}_i + (1 - \hat{\gamma}_i)\widehat{\beta_0} \tag{14}$$

where

$$\hat{\gamma}_i = \frac{\widehat{\sigma_\mu^2}}{\widehat{\sigma_\mu^2} + \widehat{\sigma^2}_i^{\text{acs}}} \tag{15}$$

This produces the Empirical best linear unbiased predictor (EBLUPs) for the given area. For models with auxillary covariates, $\widehat{\beta_0}$ is replaced by the models fitted value. Estimation of the variation of the random effects $\widehat{\sigma_\mu^2}$ is done using restricted maximum likelihood (REML).

For the frequentist intercept-only Fay-Herriot model, the package `emdi` was used. The model fitting code snippet can be found in the appendix at Listing 1. This fit specifies no correlation of the domains and provides no additional covariates.

3.2. Bayesian Fay-Herriot Model

Moving to a bayesian structure requires collecting the various distributional components. This is a hierarchical model, so there a few levels to go through. From Equation 12, it is implied that

$$Y_i \mid \theta_i, \sigma_i \sim N\left(\theta_i, \sigma_i^2\right) \tag{16}$$

From this, the conditional probability density function is

$$p(Y_i \mid \theta_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(Y_i - \theta_i)^2}{2\sigma_i^2}\right) \tag{17}$$

From this, assuming independent ε_i, the data distribution is simply the joint distribution of all locations:

$$p(\boldsymbol{Y} \mid \boldsymbol{\theta}, \boldsymbol{\sigma}) = \prod_{i=1}^{n} p(Y_i \mid \theta_i, \sigma_i) \tag{18}$$

where

$$\begin{aligned} \boldsymbol{Y} &= (Y_1, Y_2, \ldots, Y_n) \\ \boldsymbol{\theta} &= (\theta_1, \theta_2, \ldots, \theta_n) \\ \boldsymbol{\sigma} &= (\sigma_1, \sigma_2, \ldots, \sigma_n) \end{aligned} \tag{19}$$

A similar approach can be used to get the joint distribution of $\boldsymbol{\theta}$. Referring back to Equation 13, the intercept only model can be written as

$$\theta_i \mid \beta_0, \sigma_\mu = \frac{1}{\sqrt{2\pi}\sigma_\mu} \exp\left(-\frac{(\theta_i - \beta_0)^2}{2\sigma_\mu^2}\right) \tag{20}$$

and, with the assumption of independent county level effects, a joint distribution of

$$p(\boldsymbol{\theta} \mid \beta_0, \sigma_\mu) = \prod_{i=1}^{n} p(\theta_i \mid \beta_0, \sigma_\mu) \tag{21}$$

3.2.1. Prior Distributions

For the priors of β_0 and σ_μ, they were automatically provided defaults by the `brms` package. They are as follows,

$$\begin{aligned} p(\beta_0) &= t_3(\beta_0 \mid -2.1, 2.5) \\ p(\sigma_\mu) &= t_3^+(\sigma_\mu \mid 0, 2.5) \end{aligned} \tag{22}$$

where t_3^+ represents the students t distribution with 3 degrees of freedom and a lower bound of 0, otherwise known as a Half-Student-t The full density functions of these priors are not written out for the sake of space.

3.2.2. Full Posterior Distribution

Combining everything together gives:

$$\begin{aligned} p(\theta, \beta_0, \sigma_\mu \mid \boldsymbol{Y}) &\propto p(\boldsymbol{Y} \mid \theta) p(\theta \mid \beta_0, \sigma_\mu) p(\beta_0) p(\sigma_\mu) \\ &\propto \prod_{i=1}^{n} p(Y_i \mid \theta_i, \sigma_i) \times \prod_{i=1}^{n} p(\theta_i \mid \beta_0, \sigma_\mu) \\ &\times t_3(\beta_0 \mid -2.1, 2.5) \\ &\times t_3^+(\sigma_\mu \mid 0, 2.5) \end{aligned} \tag{23}$$

3.3. Spatial Random Effects

The traditional Fay-Herriot model assumes the spatial random effects are identically distributed independent random variables from a normal distribution with an unknown variance. This assumption is often inappropriate for the task of small area estimation, where autocorrelation between neighboring areas is expected. There has been a large amount of work historically in spatial variants of the Fay-Herriot model. This project will be using an Intrinsic Conditional Auto-Regressive model (ICAR) for the spatial random effects as used by Porter et al. [11]. The ICAR component used by the `brms` package is built off the work done by Mitzi Morris [12].

To facilitate the use of this component, an adjacency matrix was created using the county TIGER shapefiles provided by the Census Bureau alongside the `spdep` package.

Let W represent the binary adjacency matrix where:

$$w_{i,j} = \begin{cases} 1 & \text{if counties i and j are adjacent and } i \neq j \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

and

$$d_i = \sum_j w_{i,j} \tag{25}$$

represent the number of neighbors for each county. For this matrix, queen contiguity is used, so counties sharing a boundary or vertex are considered adjacent. From this, define the distribution of μ_i from Equation 13 as

$$\mu_i \mid \{\mu_{i \neq j}\} \sim N\left(\sum_{i \sim j} \frac{\mu_j}{d_i}, \frac{\sigma_\mu^2}{d_i}\right), \tag{26}$$

where $i \sim j$ denotes that the regions i and j are neighbors [11]. This structure allows neighboring counties to have similar random effects adjusted by the number of neighboring counties.

The prior distribution for the standard deviation of the spatial random effects match those found in Equation 22.

$$\begin{aligned} p(\beta_0) &= t_3(\beta_0 \mid -2.1, 2.5) \\ p(\sigma_\mu) &= t_3^+(\sigma_\mu \mid 0, 2.5) \end{aligned} \tag{27}$$

3.4. Auxillary covariates

For preparing the NASS crop data layer for its role as a covariate, please refer to Section 2.2.3. Extending the Bayesian model to include covariates is straightforward. Referring to Equation 20, adding in a covariate vector $\boldsymbol{\beta}_x$ gives

$$\theta_i \mid \beta_0, \boldsymbol{\beta}_x, \sigma_\mu \sim N\left(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta}_x, \sigma_\mu^2\right) \tag{28}$$

$$\theta_i \mid \beta_0, \boldsymbol{\beta}_x, \sigma_\mu = \frac{1}{\sqrt{2\pi}\sigma_\mu} \exp\left(-\frac{\left(\theta_i - \left(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta}_x\right)\right)^2}{2\sigma_\mu^2}\right) \tag{29}$$

where, for this project, $\boldsymbol{\beta}_x$ is simply β_1. This implementation expands out to support additional covariates in the same way. A similar approach is shown in [13]. Going forward, β_1 will be used instead of $\boldsymbol{\beta}_x$.

As for priors, by default `brms` gives covariate coefficients an improper flat prior. This is a uniform prior across the entire real line. Instead, a $N(0, 1)$ prior was selected β_1. The same priors for the other components still apply, so this arrangement applies both to the models with and without the ICAR component.

$$\begin{aligned} p(\beta_0) &= t_3(\beta_0 \mid -2.1, 2.5) \\ p(\beta_1) &= N(\beta_1 \mid 0, 1) \\ p(\sigma_\mu) &= t_3^+(\sigma_\mu \mid 0, 2.5) \end{aligned} \tag{30}$$

4. Results

4.1. Summary Results

Table 10 contains all of the metrics used to gauge overall model performance. All results should be interpreted with respect to the ACS results, as those function as the baseline to beat. It is important to note that this section focuses on high level aggregated metrics. Further sections will focus more on performance at the individual estimate level.

Model	MSE	MAE	Mean SD	Interval Width	Interval Coverage	Mean CRPS
ACS	0.000503	0.01697	0.01395	0.05497	0.7374	—
Int (Freq)	0.000417	0.01586	—	—	—	—
Int (Bayes)	0.000436	0.01633	0.01329	0.05197	0.7576	0.011782
CDL	0.000438	0.01636	0.01331	0.05203	0.7576	0.011787
ICAR	0.000437	0.01644	0.01307	0.05113	0.7576	0.011786
Full Model	0.000433	0.01638	0.01306	0.05107	0.7677	0.011721

Table 10: Summary metrics for all models. "Int" refers to intercept only FH models. Int (Freq) is the frequentist model fit using `Emdi` whereas Int (Bayes) was fit using `brms`. Interval width represents the mean interval width for all counties.

4.1.1. Mean Squared Error and Mean Absolute Error

Starting with MSE and MAE, all FH models show improvement over the ACS estimates for both metrcs. The best performer here is actually the frequentist intercept only model, having the lowest value for both metrics. Though it is possible that this is genuinely the best model for MAE and MSE, it is worth considering that this could indicate incorrect estimation collection from the Bayesian models. In other words, that this is a symptom of user error. Regardless, moving on to the Bayesian models they all perform similarly. The full model technically performs the best for MSE and the CDL model performs the best for MAE. The magnitude of these differences is miniscule however, and is not evidence on its own for any of the Bayesian models being the best on these metrics alone.

4.1.2. Mean Standard Deviation and Mean Interval Width

Moving onto standard deviation and interval width, some meaningful differences begin to take form. To get the baseline values for interval width, the logit transformed estimates and corresponding standard error were utilized to create a 95% interval on the original proportion scale. This was done by using those logit values to get the endpoints of the normal interval and then using the inverse logit function to transform them onto the proportion scale. Let $\hat{\theta}_i$ represent the logit transformed estimate for a county i and $\hat{\sigma}_i$ it's corresponding standard error.

$$\begin{aligned} \text{Lower}_i &= \text{logit}^{-1}\left(\hat{\theta}_i + z_{0.025}\hat{\sigma}_i\right) \\ \text{Upper}_i &= \text{logit}^{-1}\left(\hat{\theta}_i + z_{0.975}\hat{\sigma}_i\right) \end{aligned} \tag{31}$$

From this, the baseline interval width for a given county is:

$$W_i = \text{Upper}_i - \text{Lower}_i \tag{32}$$

where $z_{0.025}$ and $z_{0.975}$ represent the 0.025 and 0.975 quantiles of the standard normal respectively.

All 4 models show improvement from that baseline on these 2 metrics, showcasing the increased precision expected from FH estimates. The two models that incorporate the ICAR component show further improvement on these metrics than the intercept only and CDL model. This shows some initial evidence that incorporating that spatial autocorrelation helps with estimate precision beyond the standard Fay-Herriot model. The frequentist model is unable to be compared on these metrics due the lack of uncertainty provided by its county estimates. This is the key upside of the Bayesian approach and provides an interesting tradeoff given the results shown so far. The Bayesian models are worth consideration even if the MSE and MAE performance is truly less than the frequentist approach as the uncertainty built into the estimates is extremely valuable for small area estimates such as these.

4.1.3. Interval Coverage

As for interval coverage, it represents the proportion of counties where the true value from the census is contained within the 95% intervals provided by the model estimates. The baseline value is 0.7374, or 73/99 counties where the ACS intervals captures the true value. All of the models show improvement here, going up to 75/99 counties for 4 of the models and 76/99 counties for the full model. So not only are the estimates more precise with smaller intervals, those intervals capture more of the counties true vacancy rates. There is a tradeoff here that will be examined in a later section. At a glance it appears more counties are being captured that previously weren't with no downsides, though as will be shown later there are counties that the ACS captured that the Bayesian models lose.

4.1.4. Continuous Ranked Probability Score (CRPS)

For one final comparison of overall model performance, the Continuous Ranked Probability Score (CRPS) was captured for every prediction. The CRPS can be thought of as a generalisation of the mean absolute error (MAE) used for probabilistic outputs [14]. It's useful as it evaluates the accuracy of the overall distribution for each prediction instead of just the point estimate.

> The Continuous Ranked Probability Score (CRPS) is a generalisation of the Mean Absolute Error (MAE) tailored to probabilistic forecasting. Unlike the MAE, which compares point predictions to observations, the CRPS evaluates the accuracy of an entire predicted probaility distribution against the observed value. It does this by comparing the empirical cumulative distribution function (CDF) of the predicted values with the step-function CDF of the true value.
> Skforecast Documentation

As for the results of this score, all 4 bayesian models are nearly identical here. The full model pulls slightly ahead, but it is difficult to ascertain if the difference here is a meaningful one. It is worth noting that the intercept only model, which had the best MAE, loses out to the full model on CRPS.

4.2. Model Diagnostics

4.2.1. Summary Diagnostics

Table 11 shows the results from some basic convergence diagnostic checks for all 4 bayesian models. All models were fit with the same settings, using 4 chains, 10,000 iterations and 5,000 warmup iterations. This gave each model 20,000 post warmup draws. Summary diagnostics were taken across all model parameters, including model coefficients, variance parameters, and county-level random effects. The ICAR models also includes parameters associated with the ICAR spatial effects. The maximums and minimums were chosen as worst case scenario diagnostics.

Model	Max $\hat{r}$	Min Bulk ESS	Med Bulk ESS	Min Tail ESS	Med Tail ESS
Intercept Only	1.0017	1893	10690	3799	12868
CDL	1.0025	2104	9414	4059	12504
ICAR	1.0008	3071	38106	6008	14228
Full Model	1.0083	798	3907	1546	8605

Table 11: Model diagnostic summary. All models have 20,000 post warmup draws.

Starting with the $\hat{r}$ values, none exceed the common threshold of 1.01. The full model gets the closest to this threshold however, landing at 1.008. As for effective sample sizes, there appear to be large differences in sampling efficiency across models. The ICAR model in particular contains a far larger median effective sample size at the bulk. The full model, yet again, has the only value of concern. It's minimum bulk ESS is only 798, which could indicate, alongside its max $\hat{r}$ value, some issues with convergence. As for the intercept only and cdl models, they appear to have very similar results.

4.2.2. Trace Plots and Paired Scatterplots

Trace and paired scatterplots were examined for all 4 baysian models using the `bayesplots` package on CRAN. For this report, only the plots from the full model are shown due to its prior concerning diagnostic results. Starting with the traceplots in Figure 10, there don't appear to be any extremely concerning trends to note. All 4 chains appear to explore the distributions for all 3 variables. The `bayesplot` package warns where there are any diverging chains and it's output determined that there were none. It is worth pointing out that upon visual inspection, the intercept and prop corn soy trace plots do appear less dense than the sdcar variable.

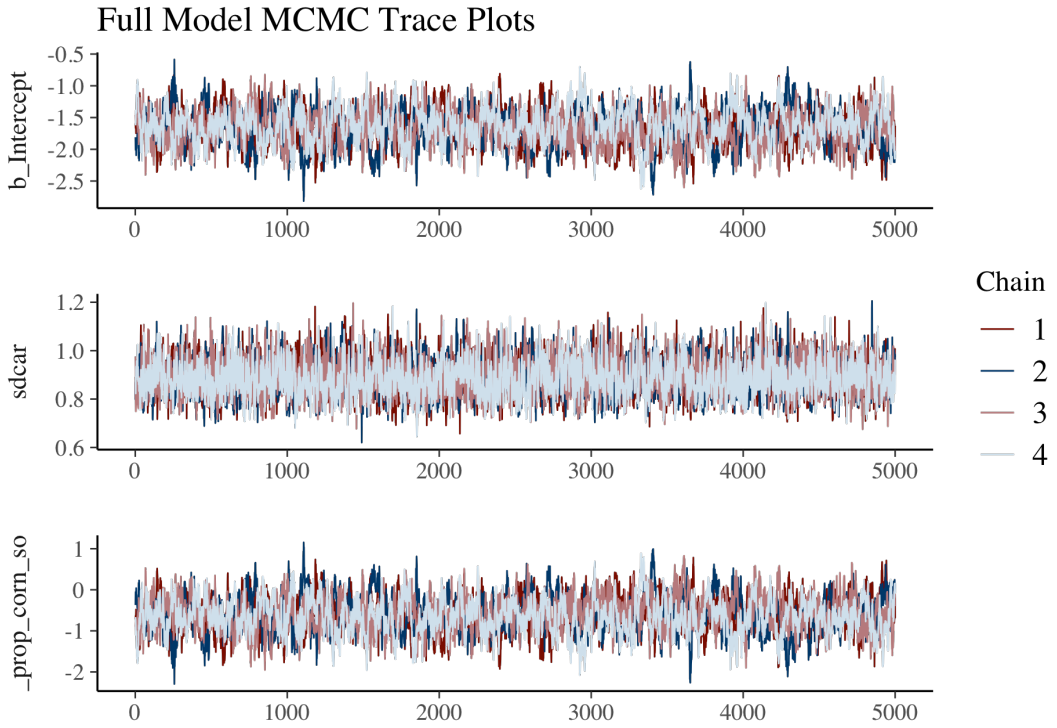


Figure 10: Trace plots from MCMC Draws for the full model (ICAR and CDL covariate).

As for the collection of plots in Figure 11, there are some interesting results. For reference, the left diagonal set of plots show the posterior histograms of each variable. The scatterplots surrounding the sdcar variable show its relationship to the other two variables. The outer-most scatterplots show the relationship between the prop corn soy variable and the intercept. The scatterplots are the most interesting to interpret. The ones surrounding sdcar indicate no apparently colinearity with the intercept or prop corn soy variables. The top right and bottom left plots however show an almost perfect negative colinearity between the intercept and prop corn soy variables. This is concerning, as it appears to indicate that the two variables are essentially sharing information and that prop corn soy is acting as a part of the intercept.

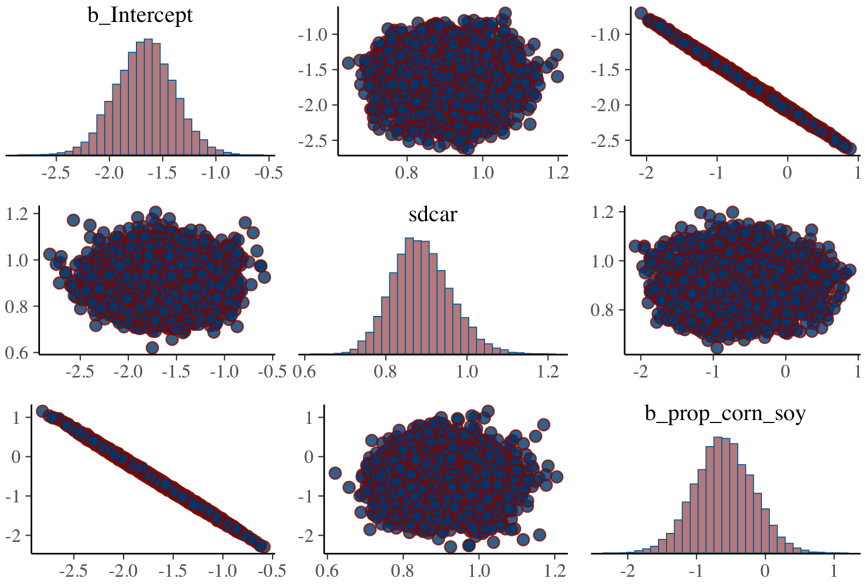


Figure 11: Variable plots from MCMC Draws for the full model (ICAR and CDL covariate).

These relationships are consistent through the various models. The ICAR model shows the same scattering as shown in the plots surrounding sdcar, and the cdl model shows the same extreme negative colinearity between variables.

4.3. Model Coefficients

Model	Variable	Estimate	SD	Lower Bound	Upper Bound
Int (Freq)	Intercept	−2.055	0.045	−2.143	−1.966
Int (Freq)	Random Effect SD	0.427	—	—	—
Int (Bayes)	Intercept	−2.056	0.046	−2.146	−1.966
Int (Bayes)	Random Effect SD	0.433	0.034	0.372	0.505
ICAR	Intercept	−2.053	0.014	−2.081	−2.025
ICAR	sdcar	0.9	0.074	0.769	1.058
CDL Only	Intercept	−2.054	0.045	−2.141	−1.967
CDL Only	Crop Proportion	−0.385	0.244	−0.861	0.101
CDL Only	Random Effect SD	0.427	0.034	0.366	0.501
Full Model	Intercept	−2.054	0.014	−2.082	−2.025
Full Model	Crop Proportion	−0.602	0.426	−1.444	0.229
Full Model	sdcar	0.888	0.072	0.758	1.039

Table 12: Coefficient distributions. Estimate for the Bayesian models represents the mean of the posterior distribution for that variable. Lower and Upper correspond to the 2.5% and 97.5% percentiles of the coefficients credible/confidence intervals respectively. Rows are colored by model to aid readability.

Table 12 provides all of the coefficients for the models fit in this project. For interpretation, it is best go variable by variable and see how they change across models. First, the intercept. This value is consistent across models, coming in around -2.005. When transformed back to the original scale gives around an $11\%(10.5\%, 12.3\%)$ vacancy rate. This makes sense given what was shown in Table 5, which showed a median rate of 10.7% and a mean of 12%. Though the estimate for the intercept is consistent across models, the standard deviation isn't. The three non-spatial models have an SD of around 0.045, whereas the two spatial models both share a value of 0.014. So, the spatial component seems to be increasing the precision of the intercept estimate.

The "Random Effects SD" variable represents the square root of the variation component of the traditional Fay-Herriot model. Within the context of the model,

$$Y_{ij} = \mu + \beta X_{ij} + U_i + \varepsilon_{ij}, \tag{33}$$

with $U_i \overset{\text{iid}}{\sim} N(0, A)$, the county random effects variable is $\sqrt{A}$. So this variable describes the distribution that all of the county random effects are realizations of. This distribution being:

$$\begin{aligned} U_i &\overset{\text{iid}}{\sim} N\left(0, 0.427^2\right) \\ U_i &\overset{\text{iid}}{\sim} N(0, 0.182). \end{aligned} \tag{34}$$

This variable is a component used to calculate the shrinkage value for each county to determine how much to alter the direct survey estimates. A promising observation from the model results is that this estimate is consistently around 0.43 across the frequentist and bayesian models, providing some evidence that the bayesian models are behaving correctly.

Another interesting observation is that the estimate barely changes when the CDL covariate is added to the model, implying that it has little effect on the between county variation. The posterior distribution of this variable from the bayesian models has a 95% credible interval of $(0.372, 0.505)$ for the intercept only and $(0.366, 0.501)$ for the CDL model.

The "sdcar" parameter, present only in the ICAR models, controls the scale of the spatial estimates for each county. This will be shown more in a later section. The estimated value for this variable is mostly consistent across the two models with it, coming in at approximately 0.90 with a credible interval of $(0.769, 1.058)$. This variable, though filling a similar role to the county random effects, cannot be directly compared in terms of pure magnitude.

As for the crop proportion variable, it builds on previous concerns. As shown in Figure 9, there is a lack of an obvious relationship between the CDL covariate and housing vacancy rates. The 95% credible intervals of this coefficient for both the CDL and Full Model lean negative but include 0, indicating that there is insufficient evidence of a non-zero effect. These distributions provide additional evidence against its inclusion. Due to this and all of the diagnostic results, the two CDL models will not be included in further result reporting.

4.4. Estimation Calculation Examples

This section was written to aid in the interpretation of the various coefficients from these models. A brief example for the county of Grundy will be shown across the three types of models. These will be simple examples meant primarily to show what the various components are doing under the hood. These estimates do not need to be calculated manually as each library has means of acquiring the estimates directly, this is purely for demonstration.

4.4.1. Intercept Only Model Frequentist

For the model fit using `emdi`, it gives a number of components that fit together to create the Fay-herriot estimate. The ones used for this example will be the intercept, variance component, the acs direct estimate and the acs estimate variance. The values are as follows:

$$\begin{aligned} \widehat{\beta_0} &= -2.055 \\ \sigma^2 &= 0.182 \\ \hat{\theta}^{\text{acs}} &= -2.502 \\ \widehat{\sigma^2}^{\text{acs}} &= 0.024 \end{aligned} \tag{35}$$

For the intercept only model, the formula for the FH estimate is:

$$\hat{\theta}^{\text{FH}} = \hat{\gamma} \cdot \hat{\theta}^{\text{acs}} + (1 - \hat{\gamma}) \cdot \widehat{\beta_0} \tag{36}$$

Where $\hat{\gamma}$, Grundy's shrinkage value, is

$$\begin{aligned} \hat{\gamma} &= \frac{\sigma^2}{\sigma^2 + \widehat{\sigma^2}^{\text{acs}}} \\ &= \frac{0.182}{0.182 + 0.024} \\ &\approx 0.883 \end{aligned} \tag{37}$$

This gives

$$\begin{aligned}\hat{\theta}^{\text{FH}} &= 0.883 \cdot -2.502 + 0.117 \cdot -2.055 \\ &\approx -2.450\end{aligned} \tag{38}$$

Which matches the FH estimate provided by emdi

```
> model_fit$ind[1, ]
  Domain    Direct        FH Out
1 Grundy -2.502095 -2.450604   0
```

For reference, this output when transformed gives:

$$\frac{1}{1+e^{-x}} = \frac{1}{1+e^{2.45}} \approx 0.0749, \tag{39}$$

or a 7.9% vacancy rate.

4.4.2. Intercept Only Model Bayesian

The approach from the `brms` output works a little bit differently. It provides an intercept, a variable representing the county effect standard error, and each county gets its own specific random effect.

They are as follows:

$$\begin{aligned}\widehat{\beta_0} &= -2.06 \\ \sigma_{\text{random effect}} &= 0.43 \\ z[\text{Grundy}] &= -0.91\end{aligned} \tag{40}$$

These components combine to get the estimated value,

$$\begin{aligned}\hat{\theta}^{\text{FH}} &= \widehat{\beta_0} + \sigma_{\text{random effect}} \cdot z[\text{Grundy}] \\ &= -2.06 + 0.43 \cdot -0.91 \\ &= -2.06 - 0.3913 \\ &\approx -2.45\end{aligned} \tag{41}$$

Which is the same as the previous example.

4.4.3. ICAR Model

As for the ICAR model, it functions identically to this one. The only difference is the standard deviation of the county effects is replaced by sdcar. For the sake of completeness, the same example will be carried out.

$$\begin{aligned}\widehat{\beta_0} &= -2.05 \\ \sigma_{\text{car}} &= 0.90 \\ z[\text{Grundy}] &= -0.44\end{aligned} \tag{42}$$

$$\begin{aligned}\hat{\theta}^{\text{FH}} &= \widehat{\beta_0} + \sigma_{\text{car}} \cdot z[\text{Grundy}] \\ &= -2.06 + 0.90 \cdot -0.44 \\ &= -2.06 - 0.39 \\ &\approx -2.44\end{aligned} \tag{43}$$

As can be seen from these examples, the estimates for Grundy barely differ at all. It is important to note that these examples do not take into account the difference in precision amongst these estimates.

4.5. Model Estimate comparisons

4.5.1. Residuals and Standard Deviations

The plots in Figure 12 show how the residuals and standard deviations change across the ACS, Intercept Only and ICAR models. Visual inspection shows a reduction in standard deviation across both models, with a more varied reduction for the ICAR model. The overall trend is the same across both models. As the magnitude of the ACS standard deviation increases, the magnitude of the reduction does as well.

For the residuals, there is more of a tossup in results. Many of the counties show a reduction in the absolute residuals for both models. For the intercept only residuals, they tend to stick a bit more closely to the reference line compared to the ICAR model. Both Fay-Herriot models show consistent reductions in the residuals for the counties with larger residual magnitudes from the ACS. Lower magnitude counties are less consistent however, with updated residuals moving in both directions.

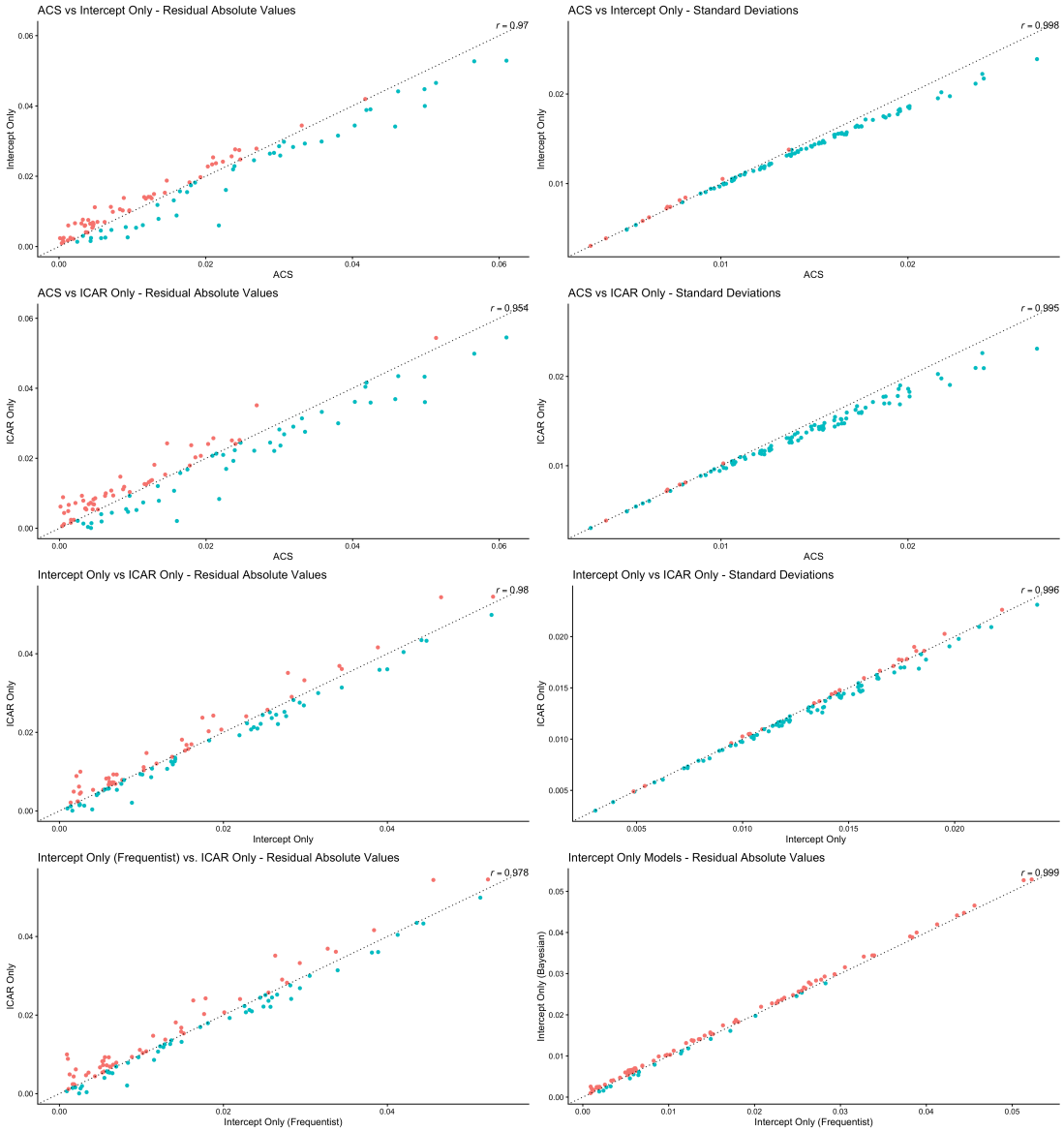


Figure 12: Plots comparing the residuals and standard deviations of vacancy rate estimates across models. Colors represent a points position relative to the reference line.

When comparing the two Fay-Herriot models directly, visual examination shows that the standard deviations are almost all lower for the ICAR model. It is worth noting that they stick extremely close to the trend line, so the difference isn't enormous. Similar behavior is seen for the residuals. It's a tossup for all of the counties, with more favoring the ICAR model. However, the difference is fairly slim and could very well just be due to random chance.

The bottom row of plots compares the Bayesian models to the Frequentist Intercept Only model. The bottom left plot with the ICAR model is, visually, nearly identical to the plot above it which examines the Bayesian Intercept Only model. The bottom right plot of the grid compares the two Intercept Only models, the Bayesian and Frequentist approaches. Almost all of the counties fall above the reference line, indicating that the Bayesian approach has slightly higher residuals than the frequentist approach across the board. They all fall very

close to the reference line however. This provides some light evidence in the direction of behavioral equivalence between Intercept Only models.

4.5.2. Interval Coverage Outcomes

When digging into improvements to residuals and standard deviations, it is always crucial to examine if the changes are meaningful and beneficial. To accomplish this, the interval coverage metric shown in Table 10 is examined more closely. Figure 13 shows counties that are gained and lost when moving to one of the two SAE models. Whether a county is captured or not is based on if the 95% prediction interval for a county contains the true value from the census.

Earlier in the report it was mentioned that the increased value of interval coverage relative to the ACS implies a gain with no tradeoff. This plot shows the tradeoff that summary metric fails to capture. The lefthand plot shows the outcomes resulting from the ICAR model prediction intervals. Of the 99 counties, 5 of them have true values that fall within the ICAR intervals that did not fall in the ACS intervals. However, 3 counties show the opposite result. These are counties captured by the ACS prediction intervals that fall outside the ICAR intervals. This results in a net gain of 2 counties.

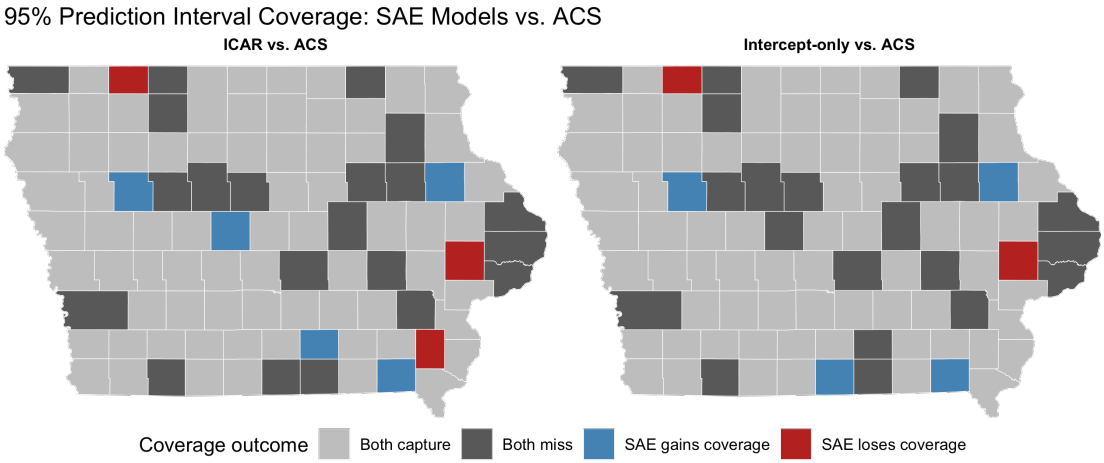


Figure 13: Plots showing the consequences of SAE estimates and intervals. Blue squares represent counties where the SAE intervals capture the true value that the ACS intervals failed to. Red is the opposite, showing true values that the ACS captured that the SAE interval failed to.

A particular limitation of the interval coverage metric appears when examining the Intercept only plot on the right. It shows the same net gain, 2 more counties gained than lost. However, that model captures 4 counties and loses 2. The interval coverage metric implies that these two models are equivalent, though they reach different conclusions across counties.

Both models are consistent across the vast majority of counties. They both gain Deleware, Sac, and Van Buren counties. They both lose Cedar and Dickinson counties. There are 4 counties where outcomes differ. Of those, the ICAR model gains Boone and Monroe county

but loses Henry county. The intercept only model gains Wayne county and does not lose Henry county.

County	Model	Estimate	True Rate	Residual	Lower	Upper	Captured
Cedar	ACS	0.0936	0.0728	0.0209	0.0724	0.1204	Yes
Cedar	Intercept	0.0961	0.0728	0.0234	0.0751	0.1209	No
Cedar	ICAR	0.0935	0.0728	0.0207	0.0729	0.1175	No
Dickinson	ACS	0.3964	0.4174	−0.0210	0.3750	0.4182	Yes
Dickinson	Intercept	0.3921	0.4174	−0.0254	0.3710	0.4138	No
Dickinson	ICAR	0.3917	0.4174	−0.0257	0.3706	0.4134	No
Henry	ACS	0.0979	0.0776	0.0203	0.0740	0.1285	Yes
Henry	Intercept	0.1004	0.0776	0.0228	0.0768	0.1288	Yes
Henry	ICAR	0.1017	0.0776	0.0241	0.0781	0.1300	No

Table 13: Results from counties that fall inside ACS prediction intervals but outside of SAE prediction intervals. Rows colored by county to aid readability.

Table 13 shows the results from those counties lost moving from the ACS to SAE models. What these results show is that, though the conclusions differ, estimates are not showing dramatic shifts. These are three counties where the true value falls at the edge of the original intervals, so any minute change results in them being lost. Them being at the edge of their respective intervals is unsurprising, as they land near the tails of the vacancy rate distribution. Dickinson is an interesting county as it has the largest vacancy rate of all counties in Iowa. There isn't a close second either, with the 2nd highest vacancy rate going to Ringgold county at 0.27. As such, a method like Fay-Herriot that brings estimates more in line with the regression results is behaving as expected. Cedar and Henry are on the opposite side, falling closer to the 1st quartile of true values (0.0789). This is well below the 1st quartile of ACS estimtates (0.0900) which explains their overestimated values.

These results are promising as they make the tradeoff discussed earlier less dramatic. The true values no longer fall within the 95% prediction interval, but the estimtates themselves barely changed. These models aren't making these county's estimates dramatically worse. These were already sensitive counties to begin with.

5. Conclusion

5.1. Limitations

This project involves multiple components and was tackled over a single semester. As such, it contains many limitations that constrain the strength of any conclusions that may be reached.

5.1.1. Spatial Covariate

The spatial covariate selected for this project is the most glaring limitation of this project. It was intended to be a showcase of the potential utility of auxiliary covariates, one of the main features of the Fay-Herriot model. However, due to circumstances outside of the project itself, the spatial covariate chosen has a largely inconsequential relationship with the response. As

such, it does a poor job of performing its role and the models with it are roughly equivalent to those without it. The cause of this was a change of topic.

The original response for this project was the proportion of workers in the agricultural industry per county, which aligned a lot more with the crop land coverage covariate. This was information only present in the ACS however and is not a topic covered by the decennial census, so there was no way to use it as a source of true values. This was a breaking oversight.

The topic was scrapped partway through the semester because of this, resulting in the housing vacancy rates explored in this report. Finding a more appropriate spatial dataset for this response was considered, though it was decided to be too time consuming for the point of the semester this occurred in. Dropping the convariate entirely was considered, but enough work was done with it to warrant its inclusion as a demonstration of potential methodologies.

It is worth noting that the crop data layer contains a ton of information, and it is possible that more sophisticated processing of its data could result in an informative covariate for housing vacancy rates. This project is also limited to Iowa at the county level in 2020, so these results do not rule out the possibility of a relationship in other areas or at other spatial/temporal scales.

5.1.2. Decennial Census as source of truth

The census is a human process and does not provide perfect numbers. The 2020 census in particular is a challenging dataset due to its overlapping occurrence with COVID-19. As briefly discussed, the U.S. Government Accountability Office documented multiple accuracy issues with the 2020 census. This was due in part to multiple challenges the Bureau was faced with, such as “late census design changes, staffing issues, and budgetary uncertainty” [2]. Iowa was not a part of the list of 14 states identified with significant population count errors, nor were their findings related to the housing data collected. However, that does not mean the housing data used in this report is free from error. This applies to every decennial census as well. None of the values provided are free from error or bias.

Using the decennial census as the source of truth for model performance also imposes limitations on the type of analysis that can be performed. As it is only performed every 10 years, comparisons made to non-census years become less reliable. This raises concerns in this project about how well these results generalize to years outside of 2020. It is extremely possible that the best model for 2020 would differ from one in 2027 for example, and this methodology provides no way to identify that drift. The Census Bureau does provide guidelines for this however, with their Population Estimates Program (PEP) that provides official estimtates for between-census years [15].

5.1.3. Neighbor matrix encoding

When addressing spatial autocorrelation there are multiple ways of encoding neighbor relationships. Only one was explored in this project, a binary neighbor encoding. That is, if a county is adjacent to another neighbor the corresponding value in its matrix is set to 1. It

is possible that other encoding methods, such as a row normalized matrix, would perform in a meaningfully different way. This was not explored due to the quantity of models already being compared. This limits the conclusions that can be drawn from the ICAR component.

5.1.4. Use of abstracted modeling packages

Many sophisticated packages were used to aid in the completion of this project. These packages, such as `brms`, `emdi` and `terra`, greatly expedite the data wrangling and modeling process. However, they also abstract out the methods and mathematics that provide the results. Conclusions were drawn under the assumption that these packages were behaving as expected, and that they were applied correctly by the user. In particular, the package `brms` functions as a wrapper for `Stan` and generates its own stan code it executes. It is possible that a manual stan implementation of the Fay-Herriot model would provide different results. That is not to say that the use of these packages was a mistake, simply that a lower level control over the results was relinquished for the sake of efficiency and that it is worth warranting.

5.2. Discussion

The Fay-Herriot model is a versatile model in the specific contexts where it can be applied. As shown in this report, with only the information present in the ACS it is capable of improving the MSE of the original direct estimates. The Bayesian implementation of the model provides further advantages, quantifying the uncertainty of each area estimate and, by extension, improvements to estimate precision. This kind of information is not carried through with the frequentist approach, giving it a meaningful niche in comparison to it. Incorporation of spatial autocorrelation through the ICAR component showed further improvements still to estimate precision, having the narrowest intervals out of the competing models for both the coefficients and county estimates. Though the spatial covariate shown in this report lacked a meaningful effect on the estimates, it demonstrates the relative ease with which auxillary covariates can be added to the model and shows the potential for improvements that were not realized in this project. As for the best model, it depends on the priorities of the statistician. There is substantial merit to the frequentist approach with its ease of use and small size. The bayesian approaches are more computationally expensive, but are worth it if there is desire for uncertainty quantification for area estimates.

Bibliography

[1] U.S. Census Bureau, “About the Decennial Census of Population and Housing.” Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/decennial-census/about.html

[2] U.S. Government Accountability Office, “2020 Census: Coverage Errors and Challenges Inform 2030 Plans.” Accessed: Oct. 18, 2026. [Online]. Available: https://www.gao.gov/products/gao-25-107160

[3] U.S. Census Bureau, "About the American Community Survey." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about/acs-and-census.html

[4] U.S. Census Bureau, "The Importance of the American Community Survey and the Decennial Census." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about.html

[5] U.S. Census Bureau, "Top Questions About the Suvey." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/about/top-questions-about-the-survey.html

[6] U.S. Census Bureau, "Using 1-Year or 5-Year American Community Survey Data." Accessed: Oct. 18, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/guidance/estimates.html

[7] USDA National Agricultural Statistics Service, "Cropland Data Layer." Accessed: Oct. 18, 2026. [Online]. Available: https://nassgeodata.gmu.edu/CropScape/

[8] U.S. Census Bureau, "Instructions for Applying Statistical Testing to American Community Survey Data." Accessed: Oct. 29, 2026. [Online]. Available: https://www2.census.gov/programs-surveys/acs/tech_docs/statistical_testing/2021_Instructions_for_Stat_Testing_ACS.pdf

[9] S. Cucu, A. Muhammad, S. M. Rijalus, C. A. Putri, and G. P. D. Sohibien, "Improving Precision in Small Area Proportion Estimation Using Logit Transformation: A Case of Internet Utilization in Papua's Regencies, Indonesia (2021)," *International Journal of Advances in Data and Information Systems*, vol. 7, no. 1, pp. 12–23, 2026.

[10] A. C. Staff, "Iowa: 2020 Census." Accessed: Oct. 29, 2026. [Online]. Available: https://www.census.gov/library/stories/state-by-state/iowa.html

[11] A. T. Porter, S. H. Holan, C. K. Wikle, and N. Cressie, "Spatial Fay–Herriot models for small area estimation with functional covariates," *Spatial Statistics*, vol. 10, pp. 27–42, 2014, doi: https://doi.org/10.1016/j.spasta.2014.07.001.

[12] M. Morris, "Spatial Models in Stan: Intrinsic Auto-Regressive Models for Areal Data." Accessed: Nov. 12, 2026. [Online]. Available: https://mc-stan.org/learn-stan/case-studies/icar_stan.html

[13] H. C. Chung and G. S. Datta, "Bayesian spatial models for estimating means of sampled and no-sampled small areas," *Survey Methodology*, vol. 48, no. 2, pp. 463–489, 2022, [Online]. Available: https://www150.statcan.gc.ca/n1/pub/12-001-x/2022002/article/00012-eng.pdf

[14] Skforecast, "Continuous Ranked Probability Score (CRPS) in probabilistic forecasting." Accessed: Nov. 06, 2026. [Online]. Available: https://skforecast.org/0.18.0/faq/probabilistic-forecasting-crps-score.html

[15] U.S. Census Bureau, "Comparing ACS Data." Accessed: Nov. 11, 2026. [Online]. Available: https://www.census.gov/programs-surveys/acs/guidance/comparing-acs-data.html#accordion-b375705b8c-item-34b0989225

6. Appendix

6.1. EMDI and BRMS Code

6.1.1. Emdi Intercept Only

```
model_fit <- emdi::fh(
    fixed = est_logit ~ 1,
    domains="name",
    vardir = "var_logit",
    combined_data=df,
    correlation="no",
    maxit=10000,
    tol=0.0001 # default
)
```

6.1.2. BRMS Intercept Only

```
fit <- brms::brm(
    est_logit | se(sd_logit, sigma=FALSE) ~ 1 + (1 | name),
    data=acs_df,
    family=gaussian(),
    seed=100,
    iter=10000,
    warmup=5000,
    save_pars = brms::save_pars(all=TRUE)
)
```

6.1.3. BRMS ICAR

```
  fit <- brms::brm(
    est_logit | se(sd_logit, sigma=FALSE) ~ 1 + car(M, type="icar", gr=name),
    data=acs_df,
    data2=list(M=neighbor_matrix),
    family=gaussian(),
    seed=100,
    iter=10000,
    warmup=5000,
    save_pars = brms::save_pars(all=TRUE)
)
```

6.2. STAN Code

6.2.1. BRMS Intercept Only STAN Code

```
> fit$model
// generated with brms 2.23.0
functions {
```

```
}
data {
  int<lower=1> N;  // total number of observations
  vector[N] Y;  // response variable
  vector<lower=0>[N] se;  // known sampling error
  // data for group-level effects of ID 1
  int<lower=1> N_1;  // number of grouping levels
  int<lower=1> M_1;  // number of coefficients per level
  array[N] int<lower=1> J_1;  // grouping indicator per observation
  // group-level predictor values
  vector[N] Z_1_1;
  int prior_only;  // should the likelihood be ignored?
}
transformed data {
  vector<lower=0>[N] se2 = square(se);
}
parameters {
  real Intercept;  // temporary intercept for centered predictors
  vector<lower=0>[M_1] sd_1;  // group-level standard deviations
  array[M_1] vector[N_1] z_1;  // standardized group-level effects
}
transformed parameters {
  real sigma = 0;  // dispersion parameter
  vector[N_1] r_1_1;  // actual group-level effects
  // prior contributions to the log posterior
  real lprior = 0;
  r_1_1 = (sd_1[1] * (z_1[1]));
  lprior += student_t_lpdf(Intercept | 3, -2.1, 2.5);
  lprior += student_t_lpdf(sd_1 | 3, 0, 2.5)
    - 1 * student_t_lccdf(0 | 3, 0, 2.5);
}
model {
  // likelihood including constants
  if (!prior_only) {
    // initialize linear predictor term
    vector[N] mu = rep_vector(0.0, N);
    mu += Intercept;
    for (n in 1:N) {
      // add more terms to the linear predictor
      mu[n] += r_1_1[J_1[n]] * Z_1_1[n];
    }
    target += normal_lpdf(Y | mu, se);
  }
  // priors including constants
  target += lprior;
```

```
  target += std_normal_lpdf(z_1[1]);
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = Intercept;
}
```

6.2.2. BRMS ICAR STAN Code

```
> icar_only$model
// generated with brms 2.23.0
functions {
}
data {
  int<lower=1> N;  // total number of observations
  vector[N] Y;  // response variable
  vector<lower=0>[N] se;  // known sampling error
  // data for the CAR structure
  int<lower=1> Nloc;
  array[N] int<lower=1> Jloc;
  int<lower=0> Nedges;
  array[Nedges] int<lower=1> edges1;
  array[Nedges] int<lower=1> edges2;
  int prior_only;  // should the likelihood be ignored?
}
transformed data {
  vector<lower=0>[N] se2 = square(se);
}
parameters {
  real Intercept;  // temporary intercept for centered predictors
  real<lower=0> sdcar;  // SD of the CAR structure
  // parameters for the ICAR structure
  vector[Nloc] zcar;
}
transformed parameters {
  // scaled parameters for the ICAR structure
  vector[Nloc] rcar;
  real sigma = 0;  // dispersion parameter
  // prior contributions to the log posterior
  real lprior = 0;
  // compute scaled parameters for the ICAR structure
  rcar = zcar * sdcar;
  lprior += student_t_lpdf(Intercept | 3, -2.1, 2.5);
  lprior += student_t_lpdf(sdcar | 3, 0, 2.5)
    - 1 * student_t_lccdf(0 | 3, 0, 2.5);
}
model {
```

```
  // likelihood including constants
  if (!prior_only) {
    // initialize linear predictor term
    vector[N] mu = rep_vector(0.0, N);
    mu += Intercept;
    for (n in 1:N) {
      // add more terms to the linear predictor
      mu[n] += rcar[Jloc[n]];
    }
    target += normal_lpdf(Y | mu, se);
  }
  // priors including constants
  target += lprior;
  // improper prior on the spatial CAR component
  target += -0.5 * dot_self(zcar[edges1] - zcar[edges2]);
  // soft sum-to-zero constraint
  target += normal_lpdf(sum(zcar) | 0, 0.001 * Nloc);
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = Intercept;
}
```